

Silah

Multi-agent outbound GTM agent for the Saudi & GCC market

Problem statement

Sales teams need to research many companies before reaching out, but not every company is a good fit for a given product. Comparing and qualifying prospects manually is time-consuming, and even after a company is qualified, finding the right contact adds another step to an already slow process.

As a result, outbound prospecting in the Saudi market is largely manual: sales teams spend hours researching companies one by one, with no consistent, explainable way to decide which prospects are actually worth pursuing first.

Solution overview

Silah is a multi-agent system that automates the outbound GTM workflow end to end: it discovers and filters candidate companies, scores each one against the user's product using a transparent, equation-based formula, and generates a personalized outreach email for the best-fit companies. The system is designed around one core constraint: every score must be explainable and reproducible, not a free-form judgment made by a language model.

System architecture

Layer	Component	Role
Input	User input	Product description and target city, provided by the sales user
API & orchestration	FastAPI + CrewAI	Exposes the service and coordinates the three agents
Agent layer	Discovery, Scoring, Outreach agents	Execute the GTM pipeline (see below)
Data layer	Structured DB + vector store	Structured company fields, and embeddings of business descriptions
External	GPT-5.6 Luna (OpenAI API)	Cost-efficient LLM used for grounded outreach-message generation
External	Public data sources	Ministry of Commerce, Argaam and similar sources for company data

The user submits a product description and a target city. The city is applied as a filter before any scoring happens, so location never enters the scoring equation itself.

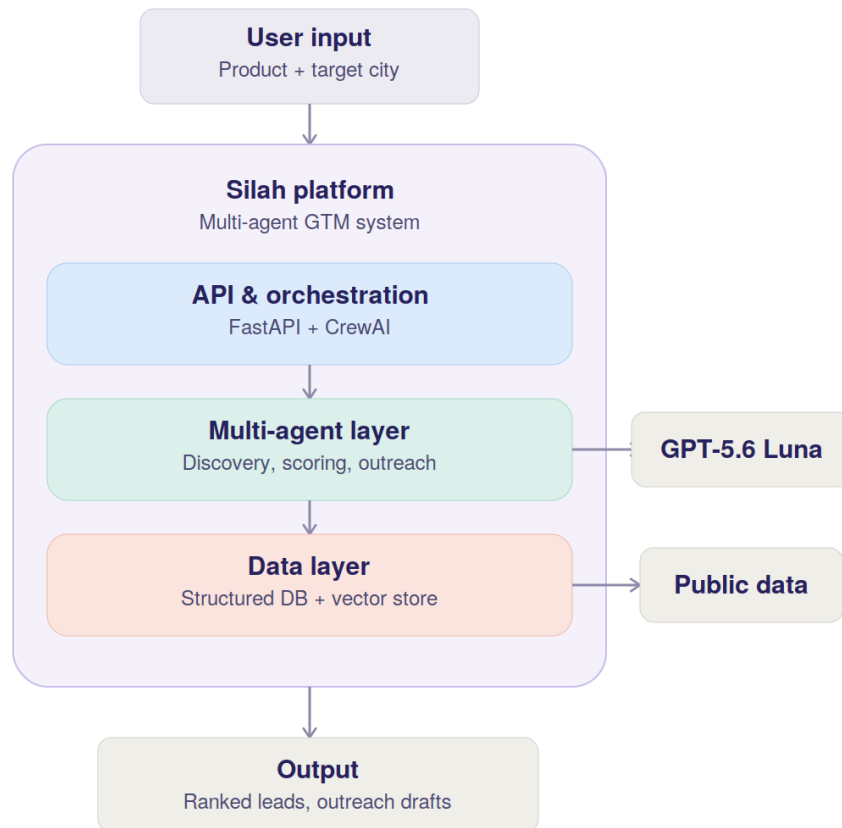


Figure 1. Silah system architecture.

AI/ML workflow

1. Data ingestion & city filtering

Company records are pulled from public sources and filtered to the city the user requested.

2. Rule-based scoring

Industry match and revenue tier are scored against fixed lookup tables (exact match / adjacent / no match).

3. Embedding-based scoring

The company's business description and the product's ideal-customer-profile (ICP) description are embedded with Sentence-Transformers; their cosine similarity is normalized with a fixed threshold scale.

4. Weighted fit score

The three scores are combined into a single fit score and companies are ranked; the top N are selected.

5. RAG-based outreach generation

For each top-ranked company, the Outreach agent retrieves its stored data and passes it to GPT-5.6 Luna (OpenAI) to generate a personalized email grounded in that data.

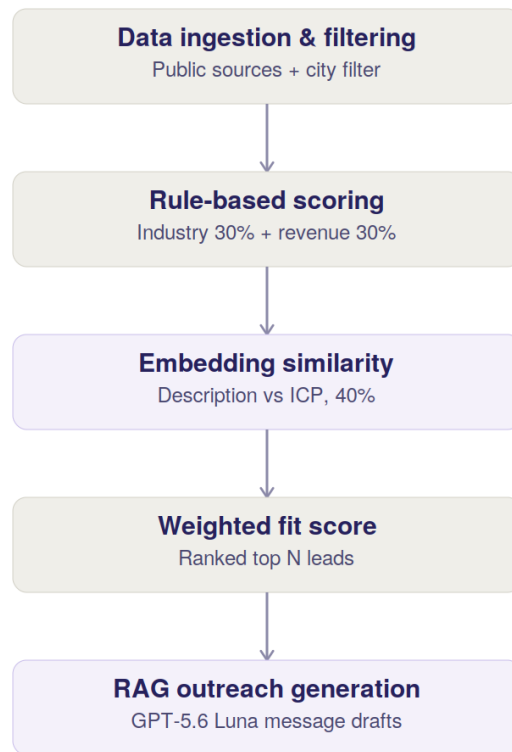


Figure 2. Silah AI/ML workflow — purple stages are ML-based (embeddings, LLM generation); gray stages are deterministic/rule-based.

Agent breakdown

Agent	Role	Technique
1. Discovery	Gathers and filters candidate companies (including the city-based filtering)	Rule-based filtering
2. Scoring	Computes a weighted fit score for every remaining company and ranks them	Company and location + embedding similarity
3. Outreach	Writes a personalized outreach email for each top-ranked company	RAG (retrieval) + GPT-5.6 Luna generation

Note the split: description similarity in Agent 2 is a single embedding-similarity calculation used as one term in the scoring equation. It is not RAG. RAG (retrieval followed by LLM generation with GPT-5.6 Luna) is used only in Agent 3, to ground each outreach email in the specific company's retrieved data.

Fit scoring equation

fit_score = (0.30 × industry_score) + (0.30 × revenue_score) + (0.40 × description_similarity_score)

Term	Weight	Source	How it's scored
Industry match	30%	Structured field	Exact match to ICP industry = 1.0, adjacent = 0.5, no match = 0
Revenue tier	30%	Structured field (public filings / Ministry of Commerce)	Target range (1.0) adjacent tier = 0.5, out of range = 0
Description similarity	40%	business_description embedding	Cosine similarity vs. ICP embedding, normalized with a fixed 0.3–0.7 range

Location is deliberately excluded from the equation: it is applied earlier as a hard filter (the user picks a target city), because most companies in the dataset cluster in a small number of major cities and would not meaningfully differentiate scores if included as a weighted term.

Data schema

Current fields: company_name, industry, location, business_description, email, website. The current dataset is mock data built from real, named Saudi companies, used to demonstrate the pipeline end to end. In production, industry, revenue and location would be sourced from verified public data (Ministry of Commerce, Argaam) rather than agent-generated estimates.

Anti-hallucination design

- Every scoring term is either a deterministic rule-based lookup (industry, revenue) or a fixed mathematical calculation (cosine similarity), never a free-form LLM judgment.
- The LLM (GPT-5.6 Luna) is only used downstream, in Agent 3, and only to write outreach text grounded in retrieved company data — not to decide who qualifies.

- Location is a hard filter set by the user, not a scored or inferred field.
- Each data point is intended to be traceable to a public source, so any score can be explained and reproduced.

Tech stack

Python, FastAPI, LangChain, CrewAI, Sentence-Transformers, OpenAI API (GPT-5.6 Luna).